

Instructor: Date:

Learner: Regional Branch:

Program: Center:

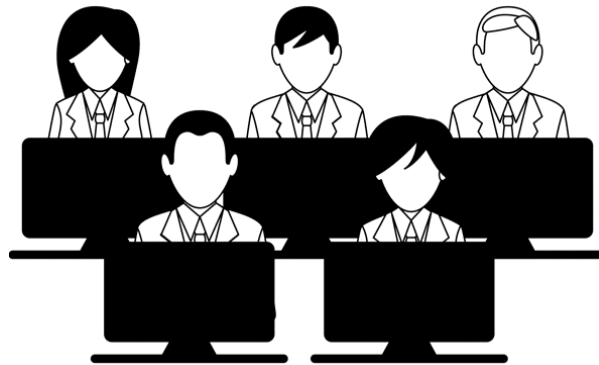
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Every new project has to meet certain specifications.



Learning Outcome:

Describe common activities at work using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can identify different types of requirements.
- I can understand why there are different types of requirements.
- I can ask questions related to different types of requirements.

Ask your partner:

- How would you define software engineering?
- What aspects does a software developer need to define before creating a computer program?





Let's Explore!

Tom is sorting the requirements of his project out*.

*To "sort something out" is to organize.



Instructions: Read the text below.

Tom: I'm not going out tonight, I have to bring this work forward.

Kelly: What are you doing?

Tom: I'm sorting the requirements of my project out.

Kelly: You aren't doing very well. You have a bunch of requirement types over there.

Tom: Well, that's because there are many types of requirements in system engineering,

Kelly: Really? Why?

Tom: Requirements encompass needs and statements of different kinds. They identify particular attributes or capabilities of a system. So, the information is massive.

Kelly: I understand.

Tom: I don't intent to narrow down the requirements into small groups because that's impossible! For now, I just need to sort out how to organize the functional, architectural, structural, and behavioral requirements; but it's complicated because there are also non-functional, performance, design, derived, allocated, and customers' requirements.

Kelly: That's a lot! What's the difference between non-functional and functional?

Tom: Well, functional requirements are used as the top-level functions. They identify tasks to be accomplished; while non-functional specifies criteria to judge not the functions, but the operation of a system.

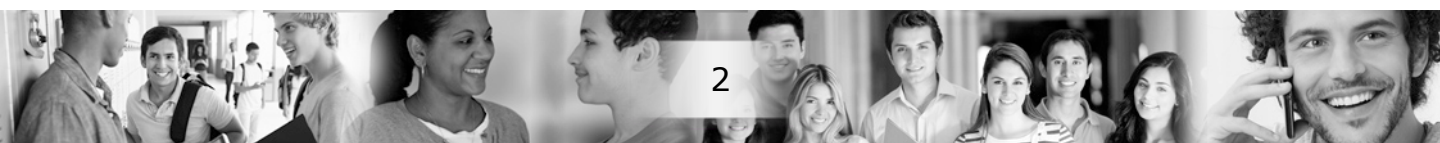
Kelly: And, what do customers' requirements do?

Tom: Those are statements of fact and assumptions that see about expectations of a system. Those expectations are mission, objectives, environment, and constraints. Customer requirements help us estimate the system's suitability and effectiveness.

Kelly: And what about the other requirements? What are they for?

Tom: Design requirements are the technical side of the system, they specify the building, coding, and executing purposes of the system. The structural requirements define the task to perform in relation to the structure of the system. And, the behavioral ones identify the system behavior or performance. Sweetie, I don't have time to talk over this now, I need to get this done.

Kelly: OK. I'm not going to interrupt you anymore. Thanks for the explanation. Bye!





Let's Practice!

Each type of requirement has different functions.

Instructions: Match Column A with Column B.

COLUMN A

1. In functional requirements,
2. To identify the user's expectations of a system,
3. A project manager needs to know...
4. A software developer has to...

COLUMN B

- a. we have to understand the customers' requirements.
- b. identify the system performance in the behavioral requirements.
- c. we need to identify the tasks to be accomplished.
- d. that the design requirements cover the technical side of the system.



Let's Practice!

Some Questions about Types of Requirements.

Instructions: Fill-in the blanks with the words in the box.

He doesn't / don't understand / I'm not following / She is not explaining

1. _____ you, so are you saying that the top-level functions have to do with the functional requirements?
2. I _____ quite well the non-functional requirements. Do they refer to the operation of the system?
3. _____ the expectations of the system. Does she know the customer's requirements?
4. _____ know anything about the design requirements. Do you want me to explain to him all the technical specifications of the system?



Extension Activity

Reviewing Information about Types of Requirements.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about types of requirements.
- Take notes about key ideas and summarize the information.
- Create a mind map about different types of requirements.
- Share your findings with your instructor and classmates.

Self-Assessment

Yes

No

- I look for more information about types of requirements.
- I take notes about key ideas.
- I summarize the information to create my mind map.
- I share my findings my instructor and classmates.

☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- **bunch** (noun): an amount of something
e.g. You have a bunch of requirement types over there.
- **constraint** (noun): a restriction
e.g. Those expectations are mission, objectives, environment, and constraints.
- **encompass** (verb): to cover or include a set of elements
e.g. Requirements encompass needs and statements of different kinds.
- **suitability** (noun): the fact of being appropriated for something
e.g. Customer requirements help us estimate the system's suitability and effectiveness.
- **top-level** (adjective): the most important aspects of something
e.g. Functional requirements are used as the top-level functions.



Learning Points

Present Simple and Present Continuous in Negative Form.

We use the **present simple tense** to talk about permanent situations such as habits or general facts. On the other hand, we use the **present continuous** when we talk about temporary facts in the present; things that are happening at the moment.

These are some of the most common uses of the **present simple** and the **present continuous**:

Present Simple

To talk about permanent situations:

*I **work** in a software company.*

To talk about routines or habits:

*A software developer always **needs** to check the users' needs.*

To talk about general truth:

*Every software package **has** to meet different requirements.*

When we need to make negative statements in the simple present, we need to add the auxiliary verbs **do** or **does** plus the particle not or we can use the contracted forms **don't** or **doesn't**:

Subject + **don't/doesn't** + **verb in base form** + complement.

Remember that we use the auxiliary **does/doesn't** for the third person singular (he, she, it) and **do/don't** for the other personal pronouns.

Examples:

*I **don't work** in a software company.*

*A software developer **doesn't need** to check the user's needs.*

*Every software package **doesn't have** to meet different requirements.*

Notice that when we use the auxiliary verbs **don't** or **doesn't**, the main verb keeps its base form. In the examples above, the verbs **work**, **need** or **have** do not change.





Learning Points

Present Continuous

To describe an action in progress:

I'm checking the software settings.

To describe a temporary situation:

She ***is doing*** a software requirement study at the moment.

When we make negative statements in the **present continuous**, we just have to add the negative word **not** next to the verb to be.

You can also use the contracted forms **isn't** and **aren't**; but remember that for the first person singular, there is not contraction of **not**:

Subject + **am not/is not/are not** + **verb in ING** + complement.

Examples:

- ***I'm not checking*** the software requirements.
- She ***is not checking*** the software requirements at the moment.
- You ***aren't recording*** the interviews.

Examples

- *I'm not going out tonight, I have to bring this work forward.*
- *You aren't doing very well, you have a bunch of requirement types over there.*
- *I don't intent to narrow down the requirements into small groups.*
- *I don't have time to talk over this now*



Learning Points

Phrasal Verbs to Solve Problems.

Phrasal verbs combine a verb with a preposition or adverb. When these words are together, they have a completely different meaning from the one they have when they are separated.

For instance, when you have the verb **get**, in the following sentence:

*She **gets** some good material to understand the software requirements.*

This verb has the meaning of collecting or obtaining something, but in the case of:

*She **got through** her work.*

The phrasal verb **get through** makes reference to finishing something.

In the following chart, we have a list of **phrasal verbs** that may be useful when you need to talk about **solving problems**:

Phrasal verbs	Meaning	Examples
Deal with	To manage a situation.	<i>We have to deal with the errors the software may have.</i>
Narrow (sth) down	To reduce something.	<i>We need to narrow the list of software errors down.</i>
Sort out Sort (sth) out	To find a solution for a situation. To organize.	<i>They have to sort out the computer program's initial problems.</i> <i>I'm sorting my ideas out.</i>
See about	To examine something or to prepare something.	<i>You should see about doing another software requirement study.</i>
Talk (sth) over	To discuss, to debate	<i>We should talk it over with the project manager.</i>

Examples

- *I don't intend to narrow down the requirements into small groups.*
- *I just need to sort out how to organize the functional, architectural, structural, and behavioral requirements*
- *Sweetie, I don't have time to talk over this now, I need to get this done.*
- *Those are statements of fact and assumptions that see about expectations of a system.*

